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Iron redox equilibrium, structure and properties of iron phosphate glasses

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Abstract

Iron redox equilibrium, structure and properties were investigated for 40Fe₂O₃–60P₂O₅ (mol%) glasses melted at different temperatures. The Fe²⁺/(Fe²⁺ + Fe³⁺) ratio increased from 17% to 50% as the melting temperature changed from 1150°C to 1400°C. The equilibrium constant, *K*, for the reaction of Fe³⁺ being reduced to Fe²⁺ varied with temperature as $\ln K = 9.40 - 1.58 \times 10^4/T$. The Raman and infrared spectra indicated that the basic iron pyrophosphate structure of the 40Fe₂O₃–60P₂O₅ (mol%) glasses did not change as the Fe²⁺/(Fe²⁺ + Fe³⁺) ratio changed. All of the properties did not change to any major degree with increasing the melting temperature. The molar volume decreased while the density increased with increasing Fe²⁺/(Fe²⁺ + Fe³⁺) ratio. It was found by DTA and XRD that two phases, Fe₃(P₂O₇)₂ and Fe₄(P₂O₇)₃, crystallized from the glass when the glass was heated in nitrogen. The crystallization behavior suggested that the amount of the crystal, Fe₃(P₂O₇)₂, may increase with increasing Fe²⁺/(Fe²⁺ + Fe³⁺) ratio, which supported the opinion that there are some structural similarities between the iron phosphate glass and the crystalline Fe₃(P₂O₇)₂ in terms of the iron coordination number and bonding of the phosphate groups. The decrease in dc resistivity and increase in dielectric constant and dielectric loss tangent, which occurred with increasing the Fe²⁺/(Fe²⁺ + Fe³⁺) ratio, were attributed to the increase of the electronic hopping from Fe²⁺ ions to Fe³⁺ ions. © 2001 Elsevier Science B.V. All rights reserved.

1. Introduction

The chemically durable iron phosphate glasses have attracted interest for the vitrification of high level nuclear wastes [1–7]. A systematic study of the compositional dependence of the structure for

binary iron phosphate glasses has been conducted by using X-ray photoelectron spectroscopy (XPS) [8], Mössbauer spectroscopy [1] and Raman spectroscopy [9].

In binary iron phosphate glasses, the [Fe³⁺] increased with increasing [Fe]/[P] molar ratio in the glass [1,8]. Meanwhile, the Fe³⁺ ions in phosphate glass acted as a network former in octahedral or distorted octahedral coordination while the Fe²⁺ ions acted as a network modifier in octahedral coordination [1,8]. With increasing iron content, the number of Fe–O–P bonds increased, while the

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number of bridging oxygen P–O–P bonds decreased [1,8] so that the phosphate network changed from the chain-like metaphosphate structure through the dimeric pyrophosphate to the isolated orthophosphate structure [9]. The good chemical durability of the iron phosphate glass was attributed to a large number of Fe–O–P bonds in the structure [1,8,9]. The 40Fe₂O₃–60P₂O₅ (mol%) glass has the best chemical durability of any binary iron phosphate glass measured to date [1].

Iron commonly exists in two valence states, as Fe²⁺ or Fe³⁺. The Fe²⁺ ⇌ Fe³⁺ equilibrium in phosphate glass depends upon the melting atmosphere and can be changed by adding different amounts of dextrose to the melt [10]. Previous work [11] has found that concentration of Fe²⁺ ions in the 40Fe₂O₃–60P₂O₅ (mol%) glass increased with increasing melting temperature and time, but the melting temperature had a stronger effect than the melting time on the relative concentration of Fe²⁺ and Fe³⁺ ions. The chemical durability of the 40Fe₂O₃–60P₂O₅ (mol%) glass was independent of the relative concentration of Fe²⁺ and Fe³⁺ ions [11].

There has been [10,12–18] much interest in the electrical and dielectric properties of iron phosphate glasses. Electrical conduction in iron phosphate glasses containing >20 mol% FeO_x was an electronic process [10]. The charge transfer due to holes (or electrons) migrating between Fe²⁺ and Fe³⁺ ions resulted in ac dispersion and dc conductivity [10,12–18]. The dielectric and mechanical relaxation processes have the same activation energy as that for dc conductivity for iron phosphate glass [17], which suggests that the same mechanism of electron hopping between iron ions is responsible for the dc conductivity and dielectric relaxation. With increasing iron content, there was a corresponding decrease in the distance between the iron ions and an increase in the dc conductivity [16].

The purpose of the present work was to determine how the iron redox equilibrium, structure and properties for an iron phosphate glass varied with the melting temperature. The changes in the phosphate network for a 40Fe₂O₃–60P₂O₅ (mol%) glass (hereafter called F40M glass) melted at

different temperatures for 1 h was investigated using Raman and infrared (IR) spectroscopy. The density, molar volume, the average distance between the iron ions and thermal expansion for this glass were also measured. Differential thermal analysis (DTA) and X-ray diffraction analysis (XRD) were used to observe the crystallization process. The dc resistivity, dielectric constant and loss tangent were also measured with changing Fe²⁺/(Fe²⁺ + Fe³⁺) ratio. The dc resistivity and loss tangent for current iron phosphate glass were compared with those for the 55FeO–45P₂O₅ (mol%) glass reported previously [10,19].

2. Experimental procedure

Batches that produced 100 g of glass were prepared by mixing reagent grade Fe₂O₃ and P₂O₅ dry crystalline powders which were melted in alumina crucibles (Coors AD998) in air at temperatures between 1150°C and 1400°C for 1 h. The melt was poured into steel molds to form 1 × 1 × 8 cm³ rectangular bars. These as-quenched glass bars were annealed at 490°C for 2 h and cooled to room temperature in 16 h. The samples of the F40M glass are designated by the melting temperature. For example, sample 1150 means that the F40M glass was melted at 1150°C.

The glass compositions listed in Table 1 are the same as those calculated previously [11] using the measured concentration of Fe²⁺ and Fe³⁺ as determined from the Mössbauer spectra, and assuming that the original Fe/P ratio (0.67) of the batch composition is retained in the glass. It was assumed [11] that there was no change in P₂O₅ content since chemical analysis has shown that the P₂O₅ content of the glass is close to that of the batch.

Samples for property measurements were cut from the annealed bars, ground, polished with 600-grit polishing paper and weighed. The density, d , of each glass was measured by the Archimedes method using deionized water as the immersion liquid. The molar volume, V_M , was calculated from the measured density of the glass and molecular weight of the glass composition. The average distance between the iron ions, R_{Fe} , was calculated from

Table 1
Composition and selected properties of the F40M glasses melted at different temperatures for 1 h

Sample	Composition (mol%) ^a		Fe ²⁺ /(Fe ²⁺ + Fe ³⁺) (%) (±0.02)	<i>d</i> (g/cm ³) (±0.02)	<i>V_M</i> (±0.3) (cm ³ /mol)	<i>R_{Fe}</i> (Å) (±0.01)	CTE ^b (10 ^{−7} /°C)	<i>T_g</i> ^c (±5°C)	<i>T_i</i> ^d (±5°C)
	FeO	Fe ₂ O ₃	P ₂ O ₅						
1150 ^e	12.7	31.1	56.2	0.17	2.95	47.0	65.4	483	518
1200	14.1	30.1	55.8	0.19	3.00	45.8	63.9	490	516
1250	20.8	25.4	53.8	0.29	3.02	43.7	67.6	484	509
1300	24.6	22.8	52.6	0.35	3.04	42.4	72.0	488	510
1350	28.8	19.8	51.4	0.42	3.08	40.7	70.7	492	516
1400	33.3	16.7	50.0	0.50	3.10	39.2	NM ^f	NM ^f	NM ^f

^a Calculated glass composition based on the Fe²⁺ fraction (Fe²⁺/(Fe²⁺ + Fe³⁺)) in the glass. Details of composition calculations have been published elsewhere [11].

^b Average coefficient of thermal expansion between 25 and 300°C, ±3.0 × 10^{−7}/°C.

^c Dilatometric glass transition temperature.

^d Dilatometric glass softening temperature.

^e Temperature in °C at which glass was melted in air for 1 h.

^f Not measured.

$$R_{\text{Fe}} = (4\pi N/3)^{-1/3}, \quad (1)$$

where *N* is the concentration of total iron ions (both Fe²⁺ and Fe³⁺ ions) per unit volume, as calculated from the glass composition and measured density. The iron ions are assumed to be uniformly distributed in Eq. (1). The estimated errors in the density, molar volume and average distance between the iron ions were estimated as ±0.02 g/cm³, ±0.3 cm³/mol and ±0.01 Å, respectively.

The coefficient of thermal expansion (CTE), the dilatometric transition temperature and dilatometric softening temperature for each glass were measured with an Orton Dilatometer (Model 1600D) on a sample of 2.54 cm in length heated in air at a rate of 3°C/min. The error in CTE, transition temperature and softening temperature is estimated at ±3.0 × 10^{−7}/°C, ±5°C and ±5°C, respectively.

The IR spectra of the glasses were measured from 450 to 1500 cm^{−1} (wave number) by the standard KBr pellet method using a fourier transform infrared (FT-IR) spectrometer (Perkin-Elmer 1760X). The pellets were prepared by mixing about 4 mg of glass powder (<25 μm) with 150 mg of dry KBr. The background for the IR spectra of the glass samples was corrected for the spectra of KBr.

Samples for the Raman measurements were parallelepipeds, approximately 0.1 cm thick cut from annealed glass bars and with an optically polished 1 × 1 cm² surface. All the Raman spectra were measured using 514.5 nm light from a 200 mW Innova 100 coherent argon ion laser and were recorded by a computerized triple monochromator, Dilor model Z24. A scattering geometry of 90° was used with the sample oriented at a near grazing angle.

The crystallization temperature for each glass was measured by DTA (DTA-7, Perkin-Elmer Corporation, Norwalk, CT, USA) at a heating rate of 10°C/min in a flowing (30 cm³/min) nitrogen atmosphere. A typical sample weight of 60 mg and the same particle size (<50 μm) glass powder were used for DTA. The measuring methods and treatment of the data were the same as reported previously [11]. The glass and its crystallization

product were identified by XRD using a Scintag XDS2000 X-ray diffractometer.

Glass samples with approximate size of $1 \times 1 \times 0.1 \text{ cm}^3$ were used for the dc electrical resistivity and dielectric property measurements at 22°C . A conductive gold film (diameter 0.6 cm) was vacuum deposited on the two $1 \times 1 \text{ cm}^2$ surfaces of the sample for the electrodes. The dc electrical resistivity was measured with a pA meter/dc voltage source (Hewlett Packard 4140B) together with a test fixture (Hewlett Packard 16055A). The capacitance, C , and dielectric loss tangent, $\tan \delta$, were measured with a multi-frequency inductance–capacitance–resistance (LCR) meter (Hewlett Packard 4274A). The relative dielectric constant, ϵ' , was calculated from the equation [20]

$$\epsilon' = C \frac{t}{0.0885A}, \quad (2)$$

where C is the measured capacitance in pF, t is the sample thickness in cm, and A is the electrode area in cm^2 . The errors in dc electrical resistivity ($\log \rho$), dielectric constant and dielectric loss tangent were estimated as ± 0.1 ($\log \Omega \text{ cm}$), ± 0.5 and ± 0.01 , respectively.

3. Results

Table 1 lists the calculated composition, based on the Fe^{2+} fraction ($\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$) in the glass, along with selected properties of each glass. Details of composition calculations have been published elsewhere [11]. All the samples in Table 1 formed glass when cast into a steel mold as shown by XRD that revealed no crystalline phases. With increased glass melting temperature (as the number in the sample identifier increased), the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ ratio increased. With increasing $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ ratio in the F40M glasses, the density increased while the molar volume decreased as shown in Fig. 1. The average distance between iron ions changed from 2.92 to $2.86 \text{ \AA}$ with increasing the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ as shown in Table 1 and Fig. 2. The CTE, T_g and T_f for these glasses were in the range of 63.9 –

$72.0(\pm 3.0) \times 10^{-7}/^\circ\text{C}$, 483 – $492(\pm 5)^\circ\text{C}$ and 509 – $518(\pm 5)^\circ\text{C}$, respectively.

Fig. 3 shows the Raman spectra of the F40M glasses melted at 1200°C , 1250°C , 1300°C and 1350°C . All the F40M glasses have similar Raman spectra. The largest intensity for the Raman shift is at 1090 cm^{-1} in Fig. 3 corresponding to dimer $(\text{P}_2\text{O}_7)^{4-}$ units, which indicates a large number of pyrophosphate groups in the network in the F40M glasses [9]. A small number of metaphosphate $(\text{PO}_3)^-$ units may also exist in the F40M glasses as indicated by the shoulder at 1250 cm^{-1} in the Raman spectra [9]. The shoulders at 923 cm^{-1} are

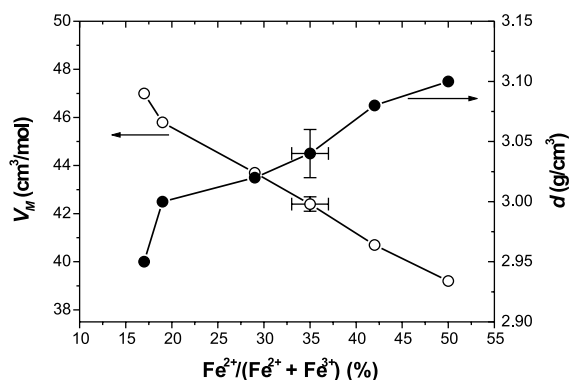


Fig. 1. Density, d (g/cm^3), and molar volume, V_M (cm^3/mol), at 22°C as a function of the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ molar ratio (%) of the F40M glasses. The lines are guides to the eye.

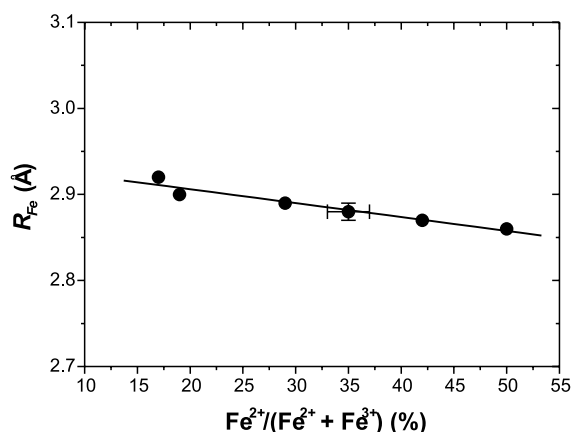


Fig. 2. The average distance between the iron ions, R_{Fe} , as a function of the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ molar ratio (%) of the F40M glasses. The line is a least-square linear fit.

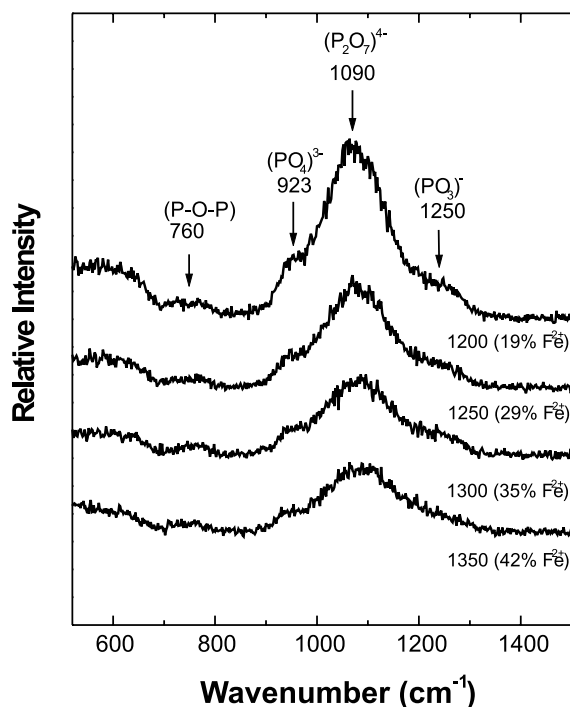


Fig. 3. Raman spectra of F40M glasses melted in air at 1200°C, 1250°C, 1300°C and 1350°C, respectively, for 1 h.

due to the vibration of the isolated $(\text{PO}_4)^{3-}$ units [9], while the vibration at 760 cm^{-1} is due to the symmetric stretch of the P–O–P bridge [9].

The IR spectra in Fig. 4 are similar for all of these glasses. There is broad absorption band centered at 1090 cm^{-1} corresponding to the asymmetric stretch of $(\text{P}_2\text{O}_7)^{4-}$ units [6,21]. The shoulder at 1280 cm^{-1} is due to $(\text{PO}_3)^-$ units and the band around 923 cm^{-1} is due to the asymmetric stretch of $(\text{PO}_4)^{3-}$ anions [5,6,21]. The band around 760 cm^{-1} is due to the symmetric stretch of P–O–P bridges [5,20]. The band around 515 cm^{-1} may be due to overlapping vibrations of iron oxygen polyhedrons and $(\text{P}_2\text{O}_7)^{4-}$ groups [5,22].

The DTA patterns (heated in nitrogen at $10^\circ\text{C}/\text{min}$) for glasses melted at 1150°C , 1250°C , 1300°C , 1350°C , and 1400°C for 1 h are shown in Fig. 5. All these DTA patterns were obtained using the same size particles, sample weight and heating rate so the data should be directly comparable [11]. The DTA curves for the glasses melted below 1300°C generally exhibited two crystallization

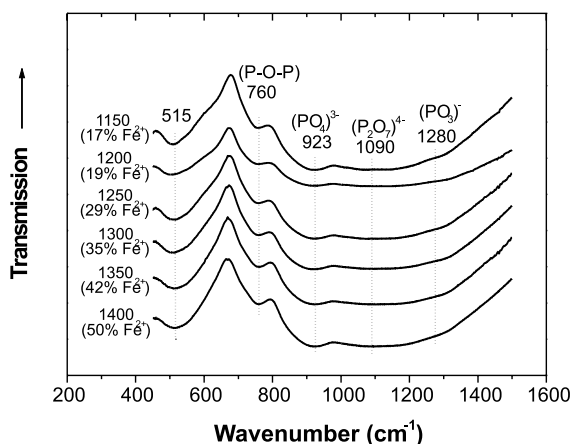


Fig. 4. The infrared spectra of F40M glasses melted in air at 1150°C , 1200°C , 1250°C , 1300°C , 1350°C and 1400°C , respectively, for 1 h.

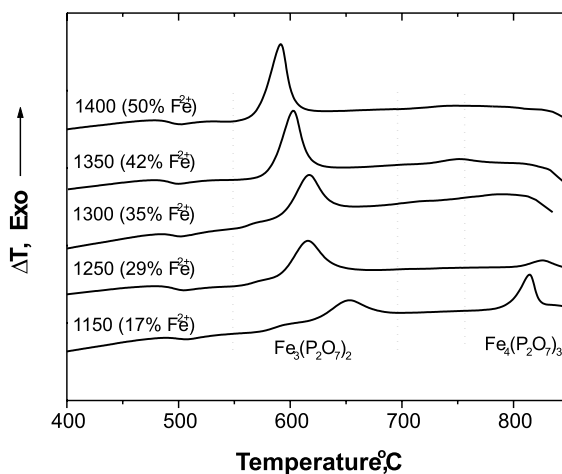


Fig. 5. DTA patterns for F40M glasses showing peaks due to crystallization of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ and $\text{Fe}_4(\text{P}_2\text{O}_7)_3$ when heated in nitrogen at $10^\circ\text{C}/\text{min}$.

peaks, but only one sharp crystallization peak at lower temperature (around 600°C) was observed for the glasses melted above 1300°C (Fig. 5). XRD analysis for the glasses crystallized in nitrogen for 24 h at the temperatures corresponding to their DTA exotherms reveals that the low temperature DTA peak is due to the crystallization of ferrous ferric pyrophosphate, $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ or $\text{Fe}^{2+}\text{Fe}^{3+}(\text{P}_2\text{O}_5)$, and the high temperature peak is due to

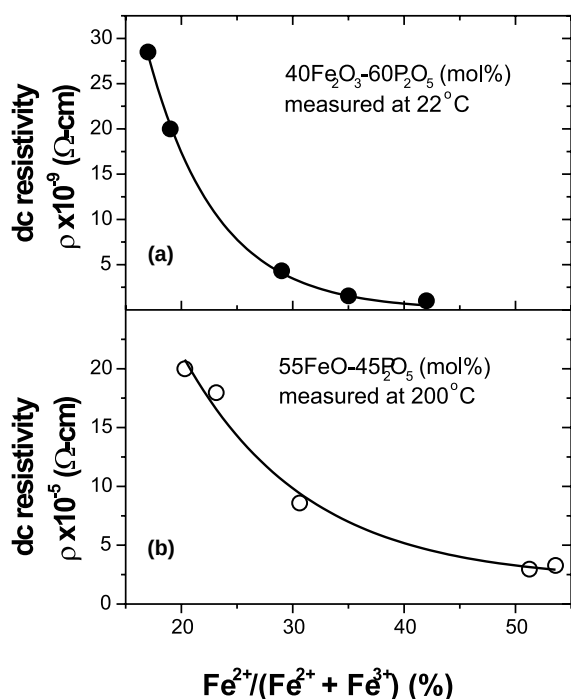


Fig. 6. The dc electrical resistivity, ρ , of the iron phosphate glasses as a function of the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ molar ratio (%). (a) For the $40\text{Fe}_2\text{O}_3\text{-}60\text{P}_2\text{O}_5$ (mol%) glass measured at 22°C ; (b) for the $55\text{FeO-}45\text{P}_2\text{O}_5$ (mol%) glass measured at 200°C [10]. Lines are the exponential fit to these glasses.

the crystallization of ferric pyrophosphate, $\text{Fe}_4(\text{P}_2\text{O}_7)_3$.

As shown in Fig. 5, the DTA peak at lower temperatures for these glasses, corresponding to the crystallization of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$, clearly shifted to lower temperatures and became larger with increasing Fe^{2+} concentration in the glass. This result suggests that the general crystallization tendency of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ for the glass increase with increasing Fe^{2+} content. The peak at higher temperatures in Fig. 5, corresponding to the crystallization of $\text{Fe}_4(\text{P}_2\text{O}_7)_3$, shifted slightly to higher temperature (from 815°C to 825°C when the melting temperature increased from 1150°C to 1250°C) and became smaller with increasing Fe^{2+} ion concentration in the glass. For glasses melted at above 1300°C , there was no peak corresponding to the crystallization of $\text{Fe}_4(\text{P}_2\text{O}_7)_3$. Apparently, the

crystallization tendency of $\text{Fe}_4(\text{P}_2\text{O}_7)_3$ decreases with increasing Fe^{2+} content.

Fig. 6 shows the dc electrical resistivity, ρ , of the F40M glasses at 22°C (a) and the $55\text{FeO-}45\text{P}_2\text{O}_5$ (mol%) glasses at 200°C [10] (b) as a function of the molar ratio of $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ in each glass. The lines in Fig. 6 are an exponential fit to the data. The dc electrical resistivity for the F40M glass melted at 1400°C , which had an $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ ratio of 0.5, was not measured.

The resistivity of the F40M glasses are in the same order with that ($7.8 \times 10^{10} \Omega \text{ cm}$) previously reported for the $40\text{Fe}_2\text{O}_3\text{-}60\text{P}_2\text{O}_5$ (mol%) glass melted at 1200°C for 2 h in air [14]. The $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ value for this glass was not reported, but it can be estimated as $\sim 20\%$ as compared to this ratio for the present F40M glass melted at 1200°C for 1 h in air, see Table 1. The resistivity of both glasses in Fig. 6 decrease exponentially with increasing the Fe^{2+} fraction in each glass up to 40% for the F40M glass and to 50% for the $55\text{FeO-}45\text{P}_2\text{O}_5$ (mol%) glasses, respectively.

Fig. 7 shows the dielectric constant, ϵ' , measured at 22°C for the F40M glasses melted at different temperatures as a function of frequency. The dielectric constant of all these glasses decreases with increasing frequency. At the same frequency, the dielectric constant increases with increasing Fe^{2+} fraction in the glass.

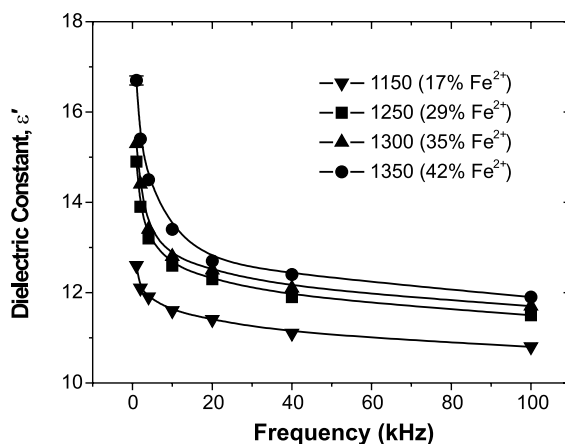


Fig. 7. The dielectric constant, ϵ' , of the F40M glasses for different frequencies at 22°C . The lines are guides to the eye.

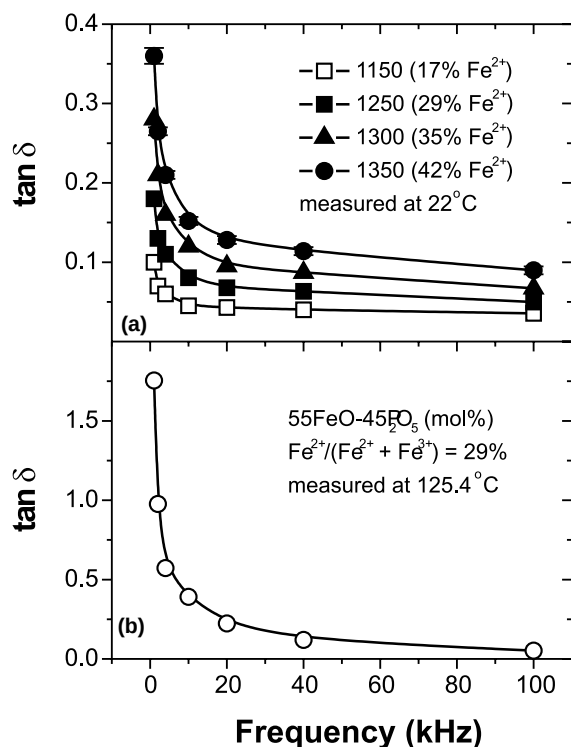


Fig. 8. The dielectric loss angle, $\tan \delta$, of the iron phosphate glasses for different frequencies. (a) For some of the F40M glasses measured at 22°C; (b) for the 55FeO–45P₂O₅ (mol%) glass measured at 125.4°C with the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ molar ratio of 29% [19]. Lines are guides to the eye.

Fig. 8 shows the loss angle, $\tan \delta$, of the F40M glasses at 22°C (a) and the 55FeO–45P₂O₅ (mol%) glass at 125.4°C with an Fe^{2+} content of 29% [19] (b) as a function of frequency. Clearly, the change in $\tan \delta$ changed with frequency is similar for both glasses. $\tan \delta$ for the F40M glass increased as the Fe^{2+} fraction increased from 17% to 42% as shown in Fig. 8(a).

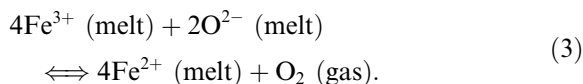
The $[\text{Fe}]/[\text{P}]$ ratio for these two glasses are close to each other (0.67 for the F40M glasses and 0.61 for the 55FeO–45P₂O₅ (mol%) glass, respectively). It is reasonable to suggest that the dc electrical resistivity and dielectric properties for both glasses may be similar when measured at the same temperature. The mechanism of electron hopping between iron ions is responsible for the dc conductivity and dielectric relaxation [17] and increase with increasing temperature [23]. Thus, it is

reasonable for the resistivity of the F40M glasses in Fig. 6(a) to be larger than that of the 55FeO–45P₂O₅ (mol%) glasses in Fig. 6(b). It is also understandable for $\tan \delta$ of the F40M glasses in Fig. 8(a) to be less than that of the 55FeO–45P₂O₅ (mol%) glasses in Fig. 8(b).

4. Discussion

4.1. Iron redox equilibrium

Iron phosphate glasses melted in air contain both Fe^{2+} and Fe^{3+} ions even when the starting batch contains only Fe^{2+} as FeO or Fe^{3+} as Fe₂O₃. The equilibrium between the iron redox couple in the melt is controlled by the following reversible reaction:



This reaction should depend upon several factors such as the melting temperature, time, and atmosphere, types of raw materials, and the presence of other substances in the batch. It was considered [11] that the iron redox equilibrium might be quickly reached within 1 h melting of 100 g batches for the iron phosphate glass. The equilibrium concentration of Fe^{2+} and Fe^{3+} ions in the melt is retained when the melt is quenched to glass. Thus, melting a binary batch composed of Fe₂O₃ and P₂O₅ results in a ternary glass composed of FeO, Fe₂O₃, and P₂O₅.

The redox equilibrium constant, K , is related to temperature by the Van't Hoff isochore [24]

$$\frac{d \ln K}{dT} = \frac{\Delta H}{RT^2}, \quad (4)$$

where ΔH is the enthalpy, R is the gas constant and T is the absolute temperature. On integration, Eq. (4) becomes

$$\ln K = -\frac{\Delta H}{RT} + I, \quad (5)$$

where I is the integration constant.

The ionic equilibrium constant K corresponding to Eq. (3) cannot be estimated due to uncertainties

in the activity of the oxygen and redox ions. However, Paul [24] indicated that when a glass of constant chemical composition is melted under a constant oxygen potential and the redox ion complexes remain unchanged within the experimental temperature range, then a plot of the ratio: $\ln[(\text{concentration of the reduced form})/(\text{concentration of the oxidized form})]$ against reciprocal absolute temperature should be a straight line, whose slope is $-\Delta H/R$.

The present data for the F40M glass agrees well with the conditions suggested by Paul [24]. Fig. 9 shows a plot of $\ln(\text{Fe}^{2+}/\text{Fe}^{3+})$ against reciprocal melting temperature ($10^4/T_M$, absolute temperature) for the F40M glass. The plot yielded nearly perfect straight line with a correlation factor of 0.991. The negative slope of the line in Fig. 9 indicates that the reaction in Eq. (3) is endothermic and will proceed toward the right-hand side to form Fe^{2+} ions with increasing temperatures. The enthalpy ΔH of this reaction, as estimated from the slope of the line in Fig. 9 by least-square fitting, is 131 ± 9 kJ/mol. So Eq. (5) can be written as

$$\ln K = 9.40 - 1.58 \times 10^4/T. \quad (6)$$

To our knowledge, this is the first time that the enthalpy ΔH of the reaction in Eq. (3) for iron phosphate glasses has been determined. Plots of $\log(\text{Fe}^{3+}/\text{Fe}^{2+})$ against reciprocal melting temperature (absolute temperature) for $30\text{R}_2\text{O}-70\text{SiO}_2$

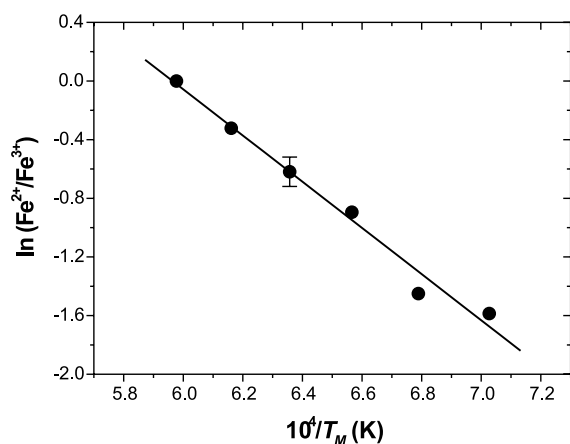


Fig. 9. Change in plot of $\ln(\text{Fe}^{2+}/\text{Fe}^{3+})$ with the reciprocal of melting temperature (K). The line is a least-square linear fit.

(mol%) glasses ($\text{R} = \text{Li}, \text{Na}, \text{and K}$) which contain ≤ 1 mol% iron have been reported previously [24]. The ΔH values to form Fe^{2+} ions (Eq. (3)) was calculated from these plots as 144 ± 5 kJ/mol, 171 ± 6 kJ/mol, and 170 ± 6 kJ/mol for $30\text{Li}_2\text{O}-70\text{SiO}_2$, $30\text{Na}_2\text{O}-70\text{SiO}_2$, and $30\text{K}_2\text{O}-70\text{SiO}_2$ (mol%) glasses, respectively. The ΔH value for the $40\text{Fe}_2\text{O}_3-60\text{P}_2\text{O}_5$ (mol%) glass is comparable to those for the alkali silicate glasses just mentioned above, suggesting that the ΔH value for Eq. (3) may be independent of the total iron content in the glass.

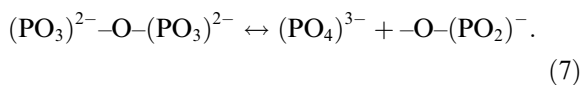
4.2. Structure

As shown in Figs. 3 and 4, respectively, the Raman and IR spectra for all the F40M glasses melted at different temperatures did not change significantly, which suggests that the structure of these glasses does not change with the melting temperature. As reported previously [11], the Mössbauer spectra indicates that the coordination environment of Fe^{2+} and Fe^{3+} ions is independent of the concentration of Fe^{2+} and Fe^{3+} ions, i.e., independent of the temperature for melting these glasses. The Fe^{2+} ions occupy sites of trigonal prism while the Fe^{3+} ions occupy sites of octahedral or distorted octahedral coordination. All the iron ions do not participate in the actual phosphate units in glass, as a result, the changes of the Fe^{2+} fraction do not affect the form of the phosphate units.

The basic phosphate network in the F40M glass is the pyrophosphate, $(\text{P}_2\text{O}_7)^{4-}$, unit since the Raman spectra revealed the largest intensity at the band of 1090 cm^{-1} (Fig. 3) and the IR spectra showed the broad absorption band centered at 1090 cm^{-1} (Fig. 4). But the phosphate network also contains some of the metaphosphate $(\text{PO}_3)^-$ and orthophosphate $(\text{PO}_4)^{3-}$ anions as evidence of such units can be found in both the Raman and IR spectra.

Tatsumisago et al. [25] studied the structure of glasses close in composition to zinc pyrophosphate ($\text{Zn}_2\text{P}_2\text{O}_7$) and suggested that along with the $(\text{P}_2\text{O}_7)^{4-}$ pyrophosphate anions, the glass contained a certain amount of isolated $(\text{PO}_4)^{3-}$ orthophosphate anions and $-\text{O}-(\text{PO}_2)^-$ fragments

(the monomer of the polymeric metaphosphate chain) due to an equilibrium between the pyrophosphate anions and the products of their disproportionation in the melt according to the following:



Such equilibrium is also believed to explain the small amount of the metaphosphate chains and isolated orthophosphate anions in the F40M iron phosphate glasses.

As indicated by DTA and XRD, the main crystalline phases formed over the temperature range from the glass transition temperature ($\sim 500^\circ\text{C}$) to the liquidus temperature ($\sim 840^\circ\text{C}$) are ferrous–ferric pyrophosphate, $\text{Fe}_3(\text{P}_2\text{O}_7)_2$, and ferric pyrophosphate, $\text{Fe}_4(\text{P}_2\text{O}_7)_3$. This crystallization behavior is evidence of the basic pyrophosphate units in the F40M glasses since the glass structure is often considered to resemble the structure of the phase crystallized from the glass.

Crystalline $\text{Fe}_4(\text{P}_2\text{O}_7)_3$ contains only Fe^{3+} ions so the amount of $\text{Fe}_4(\text{P}_2\text{O}_7)_3$ is expected to decrease in crystallized glasses as the fraction of Fe^{3+} decreases. The ratio of Fe^{2+} to Fe^{3+} ions in crystalline $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ is 1:2, so a glass containing an Fe^{2+} concentration of $\geq 33\%$ will not have any Fe^{3+} ions left to crystallize $\text{Fe}_4(\text{P}_2\text{O}_7)_3$ (second DTA peak), if all of the potential $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ fully crystallizes (first DTA peak). The absence of a second DTA crystallization peak for the glasses melted $\geq 1300^\circ\text{C}$ (Fig. 5) is consistent with their Fe^{2+} content of $\geq 35\%$ (Table 1) based on this point of view.

It should be noted that not all of the potential $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ fully crystallizes during the DTA Measurement. According to the relative Fe^{2+} and Fe^{3+} ion content in $\text{Fe}_3(\text{P}_2\text{O}_7)_2$, if the $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ fully crystallized from the glass, the amount of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ should be consistent with the FeO content in the glass (Table 1) when the glass was melted at temperature $\leq 1250^\circ\text{C}$. But the amount of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ should follow the Fe_2O_3 content (Table 1) when the glass was melted at temperature $> 1250^\circ\text{C}$. In this case the maximum amount of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ would crystallize from the glass melted

at 1300°C since its Fe^{2+} and Fe^{3+} ion content (35% and 65%, respectively) is closest to those for $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ (33% and 67%, respectively). But the magnitude of the DTA peak corresponding to the crystallization of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ in Fig. 5 indicates that the amount of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ continues to increase with the melting temperature, i.e., increasing Fe^{2+} content.

In other words, only a small portion of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ which depends on the concentration of Fe^{2+} ions, has crystallized from the glass and some amount of a glassy phase must exist after the $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ has crystallized during the anisothermal process of DTA. Thus, it is expected that the residual glassy phase may be still in a ternary $\text{FeO}-\text{Fe}_2\text{O}_3-\text{P}_2\text{O}_5$ system. Some metaphosphate and orthophosphate units may still co-exist along with pyrophosphate units in the residual glass phase since all these phosphate units are present in the F40M glass as indicated by the Raman and IR spectra in Figs. 3 and 4. The amount of Fe^{3+} ions and $(\text{P}_2\text{O}_7)^{4-}$ anions in the residual glass phase may not be enough to crystallize $\text{Fe}_4(\text{P}_2\text{O}_7)_3$ after $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ crystallized during DTA for the glasses melted above 1250°C . Such that the disappearance of the DTA peak of $\text{Fe}_4(\text{P}_2\text{O}_7)_3$ is understandable.

Neutron diffraction and Mössbauer studies suggested [26,27] that there are some structural similarities between the iron phosphate glass and crystalline, $\text{Fe}_3(\text{P}_2\text{O}_7)_2$, in terms of the iron coordination number and bonding of the phosphate groups. Crystalline $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ can be described as a network of $(\text{Fe}_3\text{O}_{12})^{16-}$ clusters, consisting of one Fe^{2+} ion in a trigonal prism and two Fe^{3+} ions in octahedral or distorted octahedral coordination. The $(\text{Fe}_3\text{O}_{12})^{16-}$ clusters are interconnected via pyrophosphate $(\text{P}_2\text{O}_7)^{4-}$ groups [26]. The crystallization behavior of the F40M glass provides the evidence for such a similarity since $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ always crystallized first from glass (first peak in DTA). It is suggested that the higher Fe^{2+} ion concentration in the glass led the glass structure to become closer to the structure of crystalline $\text{Fe}_3(\text{P}_2\text{O}_7)_2$.

According to the Raman spectra, IR spectra and DTA patterns, the structure of the F40M glass can be considered as pyrophosphate $(\text{P}_2\text{O}_7)^{4-}$ units connected with iron oxygen $[\text{FeO}_6]$ units, such as

trigonal prism $[\text{FeO}_6]$ for Fe^{2+} ions and octahedral or distorted octahedral for Fe^{3+} ions. The relative amount of trigonal prism iron anions may increase with increasing Fe^{2+} ion content as the melting temperature of the glass increased.

4.3. Properties

Since the glass structure related to the phosphate network does not seem to change much with the Fe^{2+} ion content, the general properties of the F40M glass do not change much with melting temperature. Properties related to the Fe^{2+} or Fe^{3+} ion content, such as density, molar volume, the average distance of iron ions, dc electrical resistivity and $\tan \delta$, may vary with melting temperature. However, even these properties do not change to any major degree with the melting temperature as shown in Table 1 and Figs. 1, 2, and 6–8.

It is understandable that less $[\text{O}]/[\text{Fe}]$ is present in the glass for equivalent concentrations of Fe^{2+} than for Fe^{3+} in the iron phosphate glass [8]. The results in Fig. 1 show that the density increased while the molar volume decreased with increasing the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ ratio in the F40M glass. This is because the amount of oxygen in the glass decreases as the concentration of Fe^{2+} ions increases which corresponds to the decrease in the $[\text{O}]/[\text{Fe}]$ ratio (calculated from the glass compositions listed in Table 1), see Fig. 10. Oxygen makes a major contribution to the volume of the oxygen polyhedra in the glass network. It is reasonable that the molar volume decreases when the oxygen content decreases in the glass.

It is mentioned earlier that the phosphate units do not change with the Fe^{2+} ion concentration in the glass as indicated by Raman (Fig. 3) and IR spectra (Fig. 4). Thus, the decrease in the molar volume for the glass suggests that the phosphate units in the structure may pack more densely with increasing the Fe^{2+} ions. As a result the short range structure of the iron phosphate glass tends to be close to that of crystalline $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ as discussed earlier.

It has been reported [16] that the average distance between the iron ions, R_{Fe} , becomes smaller as total iron content increases in iron phosphate glasses. The present study indicates that, when the total iron content is constant, the value of R_{Fe}

became only about 2% smaller when the Fe^{2+} ion content increased from 17% to 50%, see Table 1. This result suggests R_{Fe} is nearly independent of the relative Fe^{2+} or Fe^{3+} ion content in the glass.

The dc electrical resistivity (Fig. 6) decreased while the dielectric constant (Fig. 7) and $\tan \delta$ increased (Fig. 8) increased with increasing the relative Fe^{2+} ion content in the iron phosphate glasses from 17% to 50%. These changes in resistivity and dielectric relaxation can be explained [17] by an electron hopping mechanism between the Fe^{2+} ions and Fe^{3+} ions as opposed to the motion of ions. Increasing the Fe^{2+} ion content may strongly increase the chance for the electron jumping from Fe^{2+} ion sites to Fe^{3+} ion sites in glass.

It has been considered [16] that most of the iron ions are situated in clusters containing various numbers of Fe^{2+} or Fe^{3+} ions. Charge transport then takes place along the chain of these clusters. According to present study, the $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ can be considered as a major form of clusters in the glass. The number of $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ clusters increased while the molar volume of the glass decreased with increasing Fe^{2+} content in the F40M glass as discussed above, which may result in the denser pack of the $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ clusters in the glass. The dense pack of the cluster may lead to increase the possibility for electronic hopping of iron ions in glass.

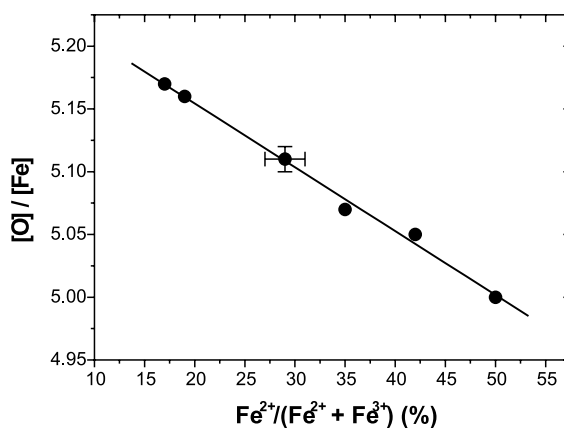


Fig. 10. The $[\text{O}]/[\text{Fe}]$ ratio in the F40M glasses as a function of the $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ molar ratio (%). $[\text{O}]/[\text{Fe}]$ ratio was calculated from the calculated glass composition listed in Table 1. The line is a least-square linear fit.

5. Conclusions

The iron redox equilibrium, structure and properties for the $40\text{Fe}_2\text{O}_3\text{--}60\text{P}_2\text{O}_5$ (mol%) glass were investigated for glasses melted in air at 1150–1400°C for 1 h. The concentration of Fe^{2+} ions increased in the $40\text{Fe}_2\text{O}_3\text{--}60\text{P}_2\text{O}_5$ (mol%) glass with increasing melting temperature. The equilibrium constant, K , for the reaction of Fe^{3+} being reduced to Fe^{2+} varied with temperature as $\ln K = 9.40 - 1.58 \times 10^4/T$. The enthalpy ΔH of the iron redox reaction to form Fe^{2+} ions for this glass was found to be 131 ± 8.7 kJ/mol.

The structure of the $40\text{Fe}_2\text{O}_3\text{--}60\text{P}_2\text{O}_5$ (mol%) glass is composed of pyrophosphate units connected with the ferrous iron trigonal prism and ferric iron octahedrals or distorted octahedrals. The structure of this glass did not change with its $\text{Fe}^{2+}/(\text{Fe}^{2+} + \text{Fe}^{3+})$ ratio. Small amounts of metaphosphate $(\text{PO}_3)^-$ and orthophosphate $(\text{PO}_4)^{3-}$ units co-exist with the pyrophosphate $(\text{P}_2\text{O}_7)^{4-}$ units in the iron phosphate structure.

Crystallization behavior discloses that the higher concentration of Fe^{2+} ions in the $40\text{Fe}_2\text{O}_3\text{--}60\text{P}_2\text{O}_5$ (mol%) glass lead the glass structure to become closer to the structure of crystalline $\text{Fe}_3(\text{P}_2\text{O}_7)_2$, which support the viewpoint that there are some structural similarities between the iron phosphate glass and the crystalline $\text{Fe}_3(\text{P}_2\text{O}_7)_2$ in terms of the iron coordination number and bonding of the phosphate groups.

The average distance between iron ions is independent of the relative Fe^{2+} or Fe^{3+} ion content in the $40\text{Fe}_2\text{O}_3\text{--}60\text{P}_2\text{O}_5$ (mol%) glass. Increasing concentration of Fe^{2+} ions may increase the possibility for electronic hopping of iron ions in this glass.

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